

# Felipe Nunes

☎ +1 (778) 839-8344 | ✉ felipenunes4822@gmail.com | 🔗 linkedin.com/in/fnunes-eng | 🐙 github.com/felipflop

## Embedded Systems Engineer

### EDUCATION

#### University of British Columbia

*Bachelor of Applied Science in Integrated Engineering*

Sep 2020 – May 2026 (Expected)

*Seeking Full-Time Positions*

### PROFESSIONAL EXPERIENCE

#### Miru Smart Technologies

May 2024 – Aug 2024

*Systems Integration Intern*

*Vancouver, B.C.*

- Reduced PCB footprint by 60% (6-layer, BGA SoM) and added a tuned 2.4 GHz inverted-F antenna
- Wrote C modules in Zephyr to drive electrochromic windows controller, increasing transitioning rate by 50%
- Devised a custom compilation pipeline using CMake and linker scripts for an ARM64 target

#### Sarcomere Dynamics

May 2023 – Apr 2024

*Electrical and Firmware Engineering Intern*

*Vancouver, B.C.*

- Implemented embedded C firmware with a finite state machine to control actuators in a multi-DOF robotic hand
- Redesigned three PCBs with emphasis on USB-C integration, mixed-signal routing, and power delivery
- Developed modular C++ classes and extended a custom Python API to support wireless communication via Wi-Fi
- Performed reflow soldering, signal probing, and hardware bring-up for embedded prototypes

### PROJECTS

#### Robotic Chessboard | Code 📄 | Report 📄

Sep 2024 – Apr 2025

- Won the Best Engineering Project award out of all final-year Capstone projects
- Designed custom PCBs and cable assemblies for a multiplexed Hall effect sensor grid to detect chess piece positions
- Selected stepper motors and drivers; configured and compiled a custom GRBL build for an H-bot gantry
- Developed hardware abstraction layers and implemented a backend algorithm for sensor polling and state detection

#### Mountable Heads Up Display | Code 📄 | Report 📄

Oct 2022 – Apr 2023

- Engineered an ESP32-based control unit to interface with UBC Formula Electric's CAN bus
- Displayed real-time telemetry (speed, temp, lap time) on an I2C OLED using FreeRTOS task scheduling
- Designed a compact custom PCB integrated into a 3D-printed helmet-mounted enclosure

#### Automated Drink Dispenser | Code 📄 | Report 📄

Nov 2021 – Apr 2022

- Constructed a custom beverage dispenser using CNC-milled aluminum and a 3D printed nozzle
- Controlled dosing system with solenoid valves and load cell feedback for precision pouring
- Wrote firmware for automated calibration, stepper control, drink selection and dispensing, and LCD user interface

### ENGINEERING DESIGN TEAMS

#### Open Robotics

Sep 2024 – Present

*Grippers Team Lead*

*University of British Columbia*

- Directed the full-cycle development of a parallel gripper end effector for the RoboCup@Home competition
- Managed a cross-disciplinary team of eight from initial design and prototyping to system integration

#### Supermileage

Sep 2022 – Aug 2023

*Electrical Team Member*

*University of British Columbia*

- Redesigned and assembled a custom PCB leveraging analog sensors to detect faults in a hydrogen electric FSAE car
- Competed and won 2023 SEMA competition in the hydrogen fuel cell category at the Indianapolis Motor Speedway

### TECHNICAL SKILLS

**Operating Systems:** Linux (Arch, Ubuntu, Raspbian), Windows, macOS, FreeRTOS, Zephyr

**Programming Languages:** C/C++/C#, Python, JavaScript, HTML, CSS, MATLAB, R, Shell Scripting, Make, CMake

**Hardware Design:** Circuit design/analysis, PCB layout (Altium, KiCad), FPGAs, VHDL, Multisim, SPICE

**Mechanical Design:** Fusion 360, Onshape, SolidWorks, Orca Slicer

**Tools & Instrumentation:** Oscilloscopes, Logic Analyzers, Function Generators, Soldering (reflow & hand)

### LANGUAGES

French (DELF B2), Portuguese (Bilingual Proficiency), Spanish (Working Proficiency)